

Vedant Somani

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EDUCATION

University of Washington, Seattle	June 2027
Master of Science - Information Management	GPA: 4.0
Acropolis Institute of Technology and Research, India	June 2025
B.Tech. - Computer Science Engineering	

WORK EXPERIENCE

Text Mining Specialist , UW Libraries & eScience Institute, (NLP, AI Research, RAG, Teaching)	Mar 2026 – Present
- Ingested 50K+ of 400K articles into Open IRE, an archiving pipeline I architected, owning the DB schema. - Recovered 6,800+ lost documents with backoff retry middleware and a dedup fix, validated by 295 e2e tests. - Improved RAG recall@5 by 14pp via sentence-anchored chunking; shipped a Haystack service cutting query time 30%.	
Fullstack Developer Intern , OS Technology Service, (React Native, Django, PostgreSQL)	Jan 2025 – Jul 2025
- Scaled a React Native and Django live-event app to 500+ concurrent users at ~200ms, lifting session length 25%. - Held 99.9% uptime and cut deploy time 40% using AWS (EC2, Fargate), CI/CD, OTA updates, and DataDog monitoring.	
Machine Learning Intern , Xenon Nation, (PyTorch, FastAPI, Docker)	Mar 2024 – Sept 2024
- Cut inference latency 35% (-250ms) via ONNX export and FastAPI response caching, deployed with Docker on AWS ECS. - Boosted sentiment F1-score 18% (74% to 87%) fine-tuning cardiffnlp/twitter-roberta-base-sentiment via custom PyTorch loop (89% precision, 85% recall).	

PROJECTS

NanoScale-LM – Language Model from Scratch , (PyTorch, Transformers, DPO, Quantization)
- Beat GPT-2 in-domain (0.55 vs 0.94 bits/byte) with a 3.1× smaller LM built from scratch in PyTorch (GQA, RoPE, DPO). - Converged 2× faster with Muon vs AdamW (p<0.0001, 5 seeds) and shrank the model 17.7× via knowledge distillation.
Uplift Modeling for Retention Targeting , (Causal Inference, Experimentation, MLflow, Streamlit)
- Lifted campaign net value 40% over a response model on 64K customers by targeting with causal uplift (X-learner) - Captured 89% of achievable uplift on ground-truth data by benchmarking 4 CATE learners with Qini and bootstrap CIs.

SKILLS

Machine Learning & Data Systems: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Snowflake, PySpark, Apache Kafka, MATLAB, Supervised Learning, Causal Inference, Hypothesis Testing, MLflow
Languages and Data: C#, C++, Python, Java, JavaScript, TypeScript, MySQL, PostgreSQL, MongoDB, Pinecone
AI & Automation: Langchain, LangGraph, LangSmith, LlamaIndex, RAG pipelines, BERT, Haystack, AutoGen, CrewAI, MCP, Tool calling, Multi Agent systems, LLM Evals, RAGAS, Phoenix, DeepEval
Platforms: AWS, Docker, Linux/Unix, Kubernetes, Git & GitHub, FastAPI, Django, React, Next.js, Node.js

ACHIEVEMENTS & PUBLICATIONS

- iEngage – Salesforce – Lovable **Hackathon Runner-up**
- Association of Information Management Students at UW (UW AIMS) **Product Competition Runner-up**
- IEEE ACROSET 2025:** PyFao-Driven Precision Irrigation and Fertilizer Optimization in Agriculture
- AWS Academy Graduate** – AWS Academy Cloud Architecting